

EN8 Steel Datasheet

EN8 Steel (also known as 080M40) is a medium strength steel with good tensile strength. It is widely used in automotive and general engineering applications. This datasheet covers the chemical composition, mechanical properties, equivalents, heat treatment, machinability, and applications of EN8 steel.

Standards & Specifications

- BS 970: 080M40
- AISI: 1040 / 1045
- DIN: C40 / C45
- JIS: S40C / S45C
- UNS: G10400

Chemical Composition (% max)

C	Mn	Si	S	P	Fe
0.36–0.44	0.60–1.00	0.10–0.40	0.05 max	0.05 max	Balance

Mechanical Properties

Tensile Strength	550–850 MPa
Yield Strength	≥ 280 MPa
Elongation	16–18%
Hardness	149–207 HB
Impact Strength (Izod)	≥ 27 J

Heat Treatment

- Normalizing: 830–860°C
- Annealing: 650–700°C
- Hardening: 820–860°C, oil or water quench
- Tempering: 550–660°C (depending on hardness requirement)

Machinability & Weldability

Machinability: EN8 has good machinability in normalized condition.

Weldability: Welding is possible with proper pre-heating and post-weld heat treatment. It is generally considered to have fair weldability.

Equivalent Grades

AISI	1040 / 1045
DIN	C40 / C45
JIS	S40C / S45C
UNS	G10400

BS	080M40
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Applications

- Automotive crankshafts
- Axles and spindles
- Studs, bolts, and general-purpose machine parts
- Shafts and gears
- General engineering components requiring medium strength

Surface Treatments

- Carburizing
- Induction hardening
- Nitriding

Comparison: EN8 vs EN19

Property	EN8	EN19
Tensile Strength	550–850 MPa	850–1000 MPa
Hardness	149–207 HB	197–255 HB
Applications	General mechanical parts	High-strength shafts, gears

Notes

EN8 Steel (080M40) is one of the most widely used medium carbon steels worldwide. This datasheet provides details on chemical composition, mechanical properties, heat treatment, equivalents, and applications. EN8 is often compared with EN19 for applications requiring higher strength.